Set notation

1

a $S = \{1, 2, 3, 4, 5, 6\}$

b $S = \{1, 3, 5, 7, 9\}$

2 $\{5, 6, 7\}$

3

a People who do not like softball.

b People who like either football, softball or swimming.

4

a Yes

b No

Intersection and union

5

a The set of all the elements in the universal set that are not in set A.

b The set containing all the elements that are members of set A or of set B or of both sets.

c The set containing all the elements that are common to both set A and set B.

6

a $\{1, 2, 4, 8, 16\}$

b $\{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17, 21\}$

7 $\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$

8

a $A \cap B = \{0, 4, 16, 36, 64\}$

b $A \cap B = \{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

c $A \cap B = \{3, 7\}$

9

a False

b False

10

a True

b True

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11

a

i $\{1,2,3,4,6,8,9,12,18,24,36\}$ **ii** $\{1,2,3,4,6,12\}$

b

i $\{1,2,3,4,5,6,7,8,9,10,11,13,17,19,23,29,31,37\}$

ii $\{2,3,5,7,11\}$

12

a $\{1,2,3,4,5,6\}$ **b** $\{1,2,3,4,5,6,7,8\}$ **c** $\{1,2,3,4,5,6\}$ **d** $\{1,2,3,4,5,6,7,8\}$

13

a

i The universal set.

ii The universal set, except for the elements of A and the elements of B.

b

i The universal set.

ii The universal set, except for the elements of A and the elements of B.

14

a The universal set, except for the elements that are in both A and B.

b The set of common multiples of 4 and 6 greater than or equal to 70.

c No

15

a The set of integers that are not multiples of 4 less than 70 or multiples of 6 less than 70.

b The set of integers that are not less than 70, and are multiples of both 4 and 6.

c No

16 Yes

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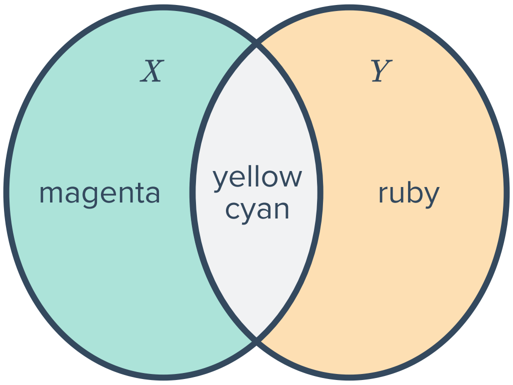
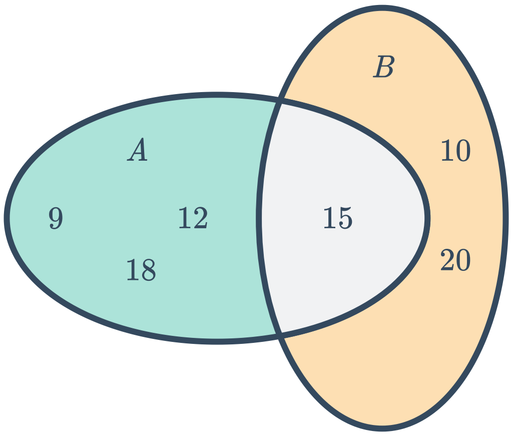
Venn diagrams

17

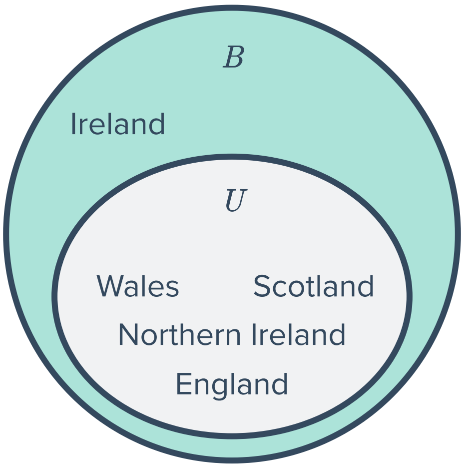
a



b



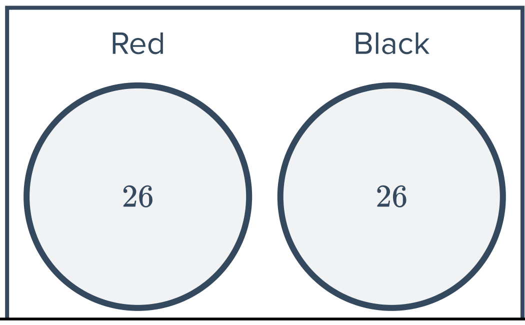
c



18 No

19 4 regions

20



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21

a $A' = \{7, 14, 12, 10\}$

b $B' = \{4, 5, 1, 12, 10\}$

22

a $A \cap B = \{7, 8\}$

b $A \cup B = \{2, 4, 6, 7, 8, 10, 14\}$

23

a $A = \{1, 3, 4, 5, 9, 10\}$

b $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

c $B' = \{1, 2, 9, 10, 11, 13\}$

24

a $A \cap B' = \{3, 4, 6\}$

b $(A \cup B)' = \{15, 60\}$

25

a $A \cap C = \{4, 5, 9\}$

b $(B \cap C)' = \{1, 2, 3, 7, 9, 10, 11, 12, 13, 14\}$

c $A \cap B \cap C = \{4, 5\}$

26

a Yes

b No

27 No

28

a $A \cap (B \cup C) = \{3, 4, 5, 9\}$

b $(A \cap B)' = \{1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

29

a 2

b 5

c 3

d 4

e 8

f 7

g 1

h 6

30

a Yes

b Yes

c The elements d and e are in both sets, so if we try to find the total number of elements in either set by adding $n(A)$ and $n(B)$, we will be counting these elements twice. Therefore, we will have to subtract the number of elements common to both sets.

31 54

32 48

33

a Venn diagram 4

b Venn diagram 3

c Venn diagram 2

d Venn diagram 1

34

a 169

b 223

c 192

d 138

e 262

f 293

35

	Play Rugby League	Don't play Rugby League
Play Rugby Union	43	185
Don't play Rugby Union	179	146

Mutually exclusive events

36 C, D

37

a

i $x=0$

ii Mutually exclusive

b

i $x=0$

ii Mutually exclusive

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38

a

i 0.3

ii 0.2

iii 0.1

iv Non-mutually exclusive

b

i 0.2

ii 0.1

iii 0

iv Mutually exclusive

c

i 0.6

ii 0.1

iii 0.2

iv Mutually exclusive

d

i 0.1

ii 0.7

iii 0

iv Mutually exclusive

39

a 0

b Yes

40 0.36

41 $\frac{3}{4}$

Non-mutually exclusive events

42 The probability of getting a $1, 3, 5$ or 6.

43 $P(A \cup B) = \frac{1}{3}$

44 $P(A \cap B) = \frac{2}{5}$

45 $P(B) = \frac{1}{5}$

46 $P(A \text{ or } B) = 0.95$

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